

ARDUINO PROJECT

Vocaloid headphones



ILKA WERTZ
2026

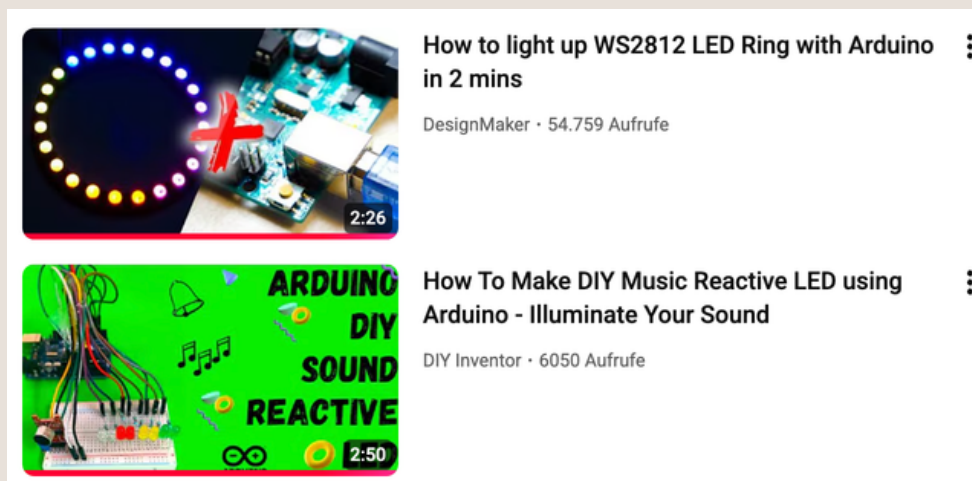
INITIAL IDEA



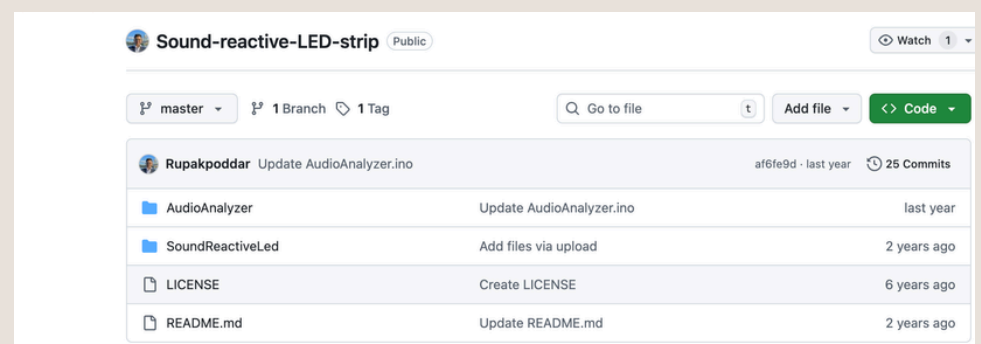
@erizadesu on TikTok

I know a lot of cosplayers use an Arduino for cosplay props, so I knew early on that I wanted to make something cosplay-related. After doing some research, I got my initial idea from this cosplayer, who made a music-reactive Hatsune Miku wig. I really liked the idea of making something sound reactive, and I thought of the headphones Miku and Luka wear on an old Vocaloid song thumbnail, as I thought it would look really cool if they could light up. Initially, I also wanted to attach an LED strip on top, but I decided against it; it looks cleaner now. Other than that, my final project is very close to my initial idea of how the headphones would look and function like.

RESEARCH



I watched a lot of videos on the WS2812 and sound-reactive LEDs in general and these were the ones I found most helpful.



Big shoutout to this GitHub user who wrote the code I ended up mostly using.

REFERENCES

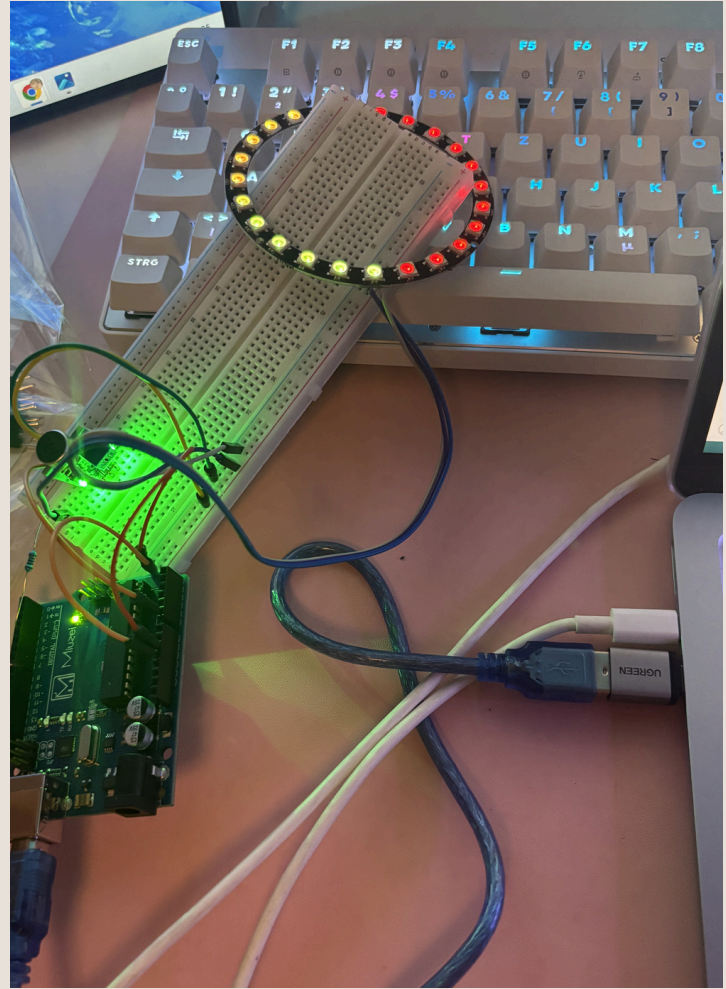
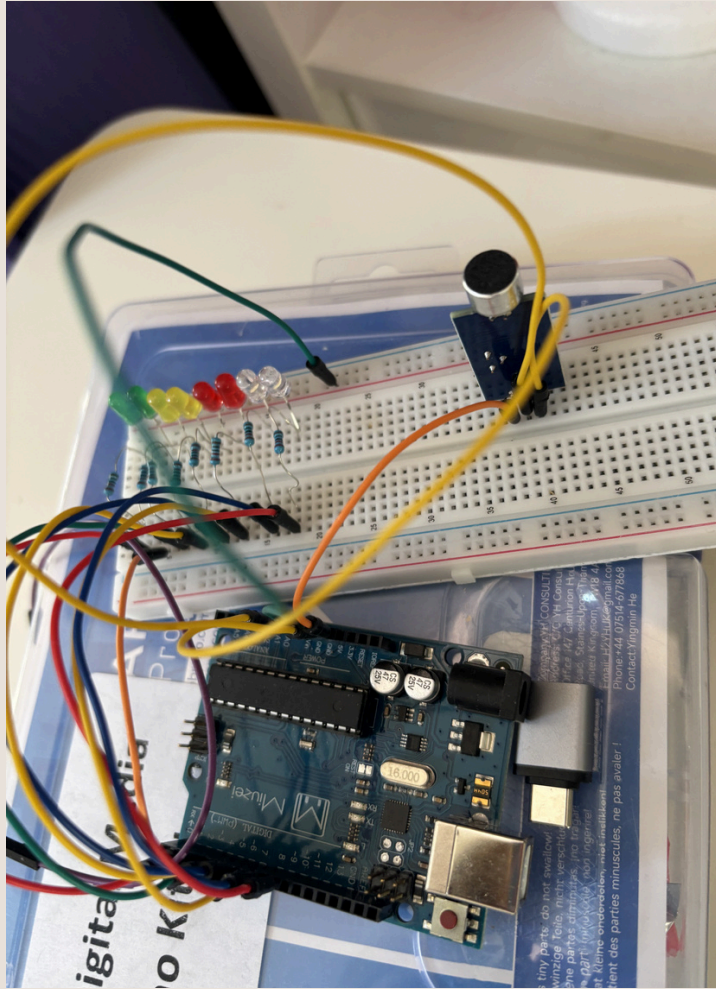


The original Cover for the song Magnet, released in 2009



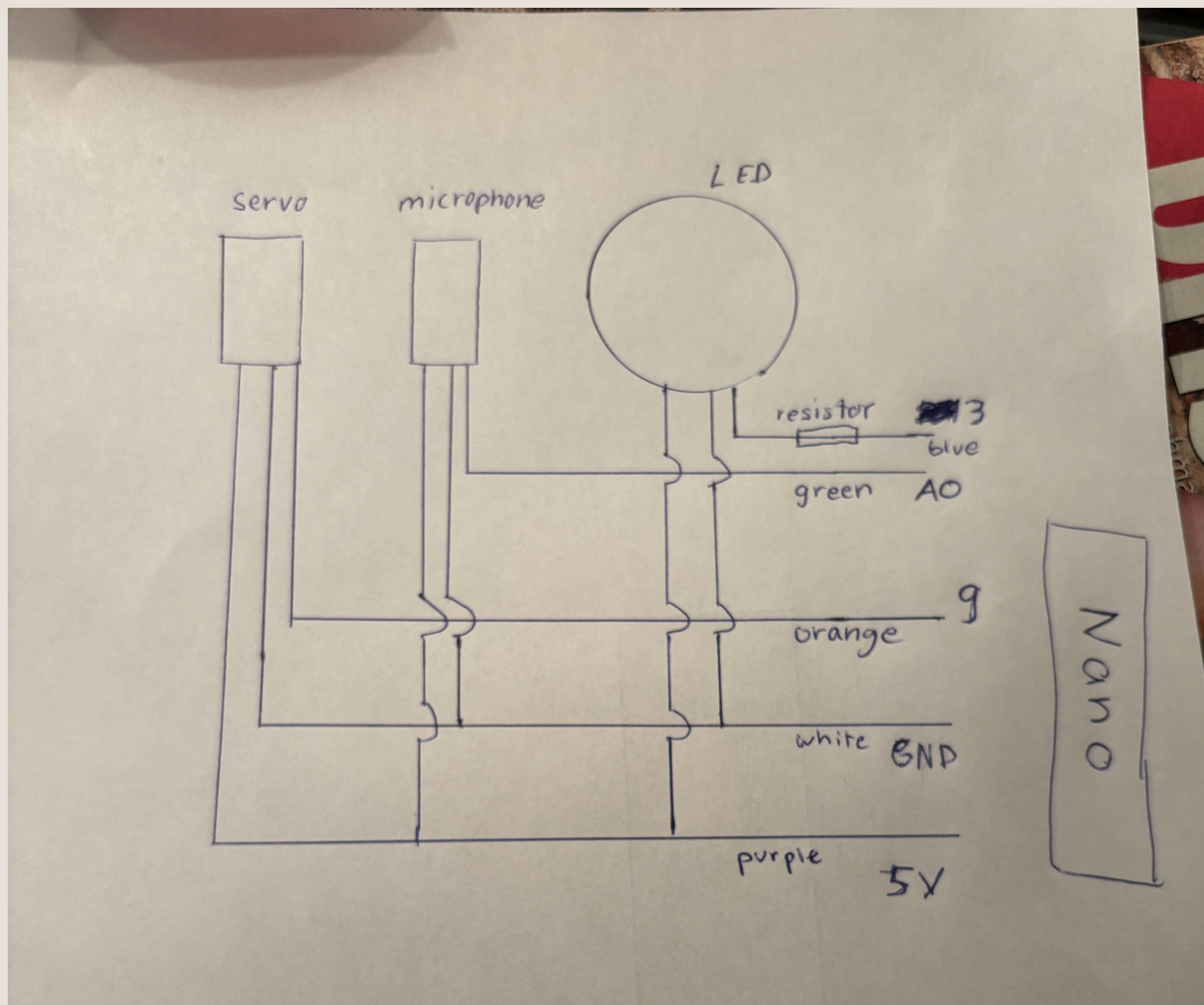
Previous interpretations of the design by other cosplayers

FIRST TESTING



Pictures from my first tests that I also presented in class.

PROCESS



Circuit diagram sketch

PROCESS

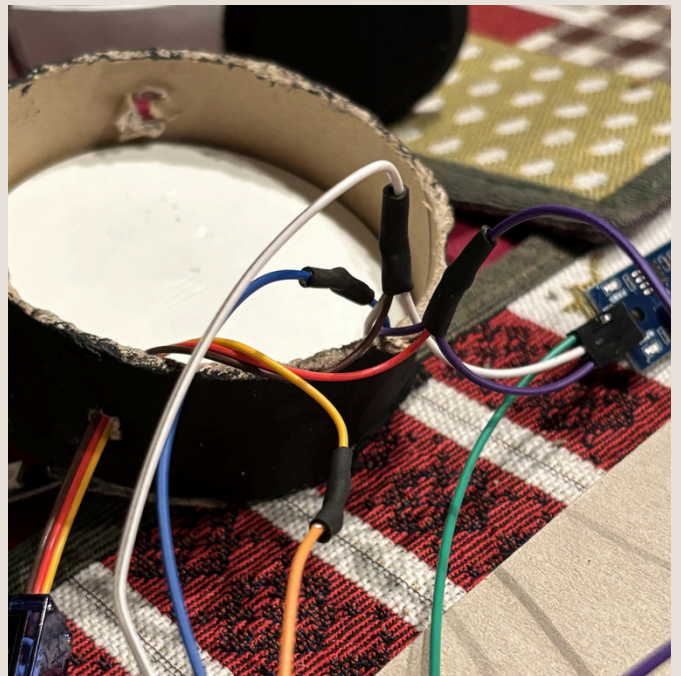
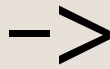
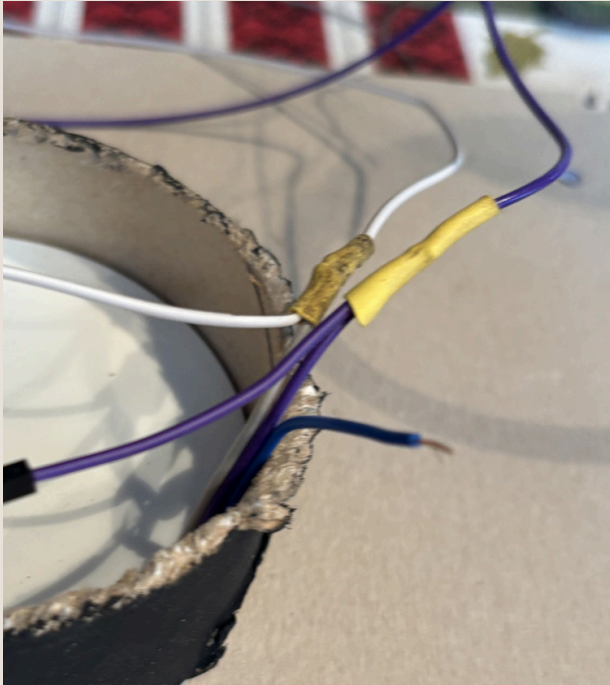


For the base I used an old pair of headphones and painted them.



After soldering the wires on the LED ring, I put hot glue over them for stability and isolation. I didn't take a better picture, but between DI and the blue cable, which connects to pin 3, is a 330 ohm resistor.

PROCESS



Heat shrinkable tubing (that I had to cut open again because I forgot about the servo...)

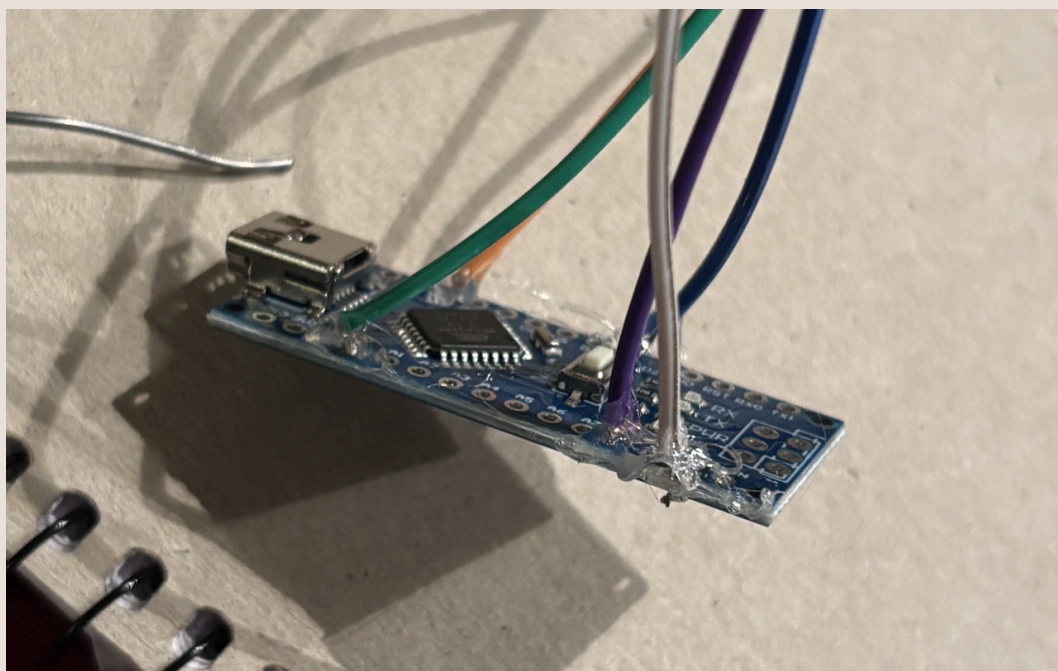


Hole for the microphone to poke out so it picks up sound better. As far away from the servo as possible so the servo noise wouldn't affect the LEDs too much.

PROCESS

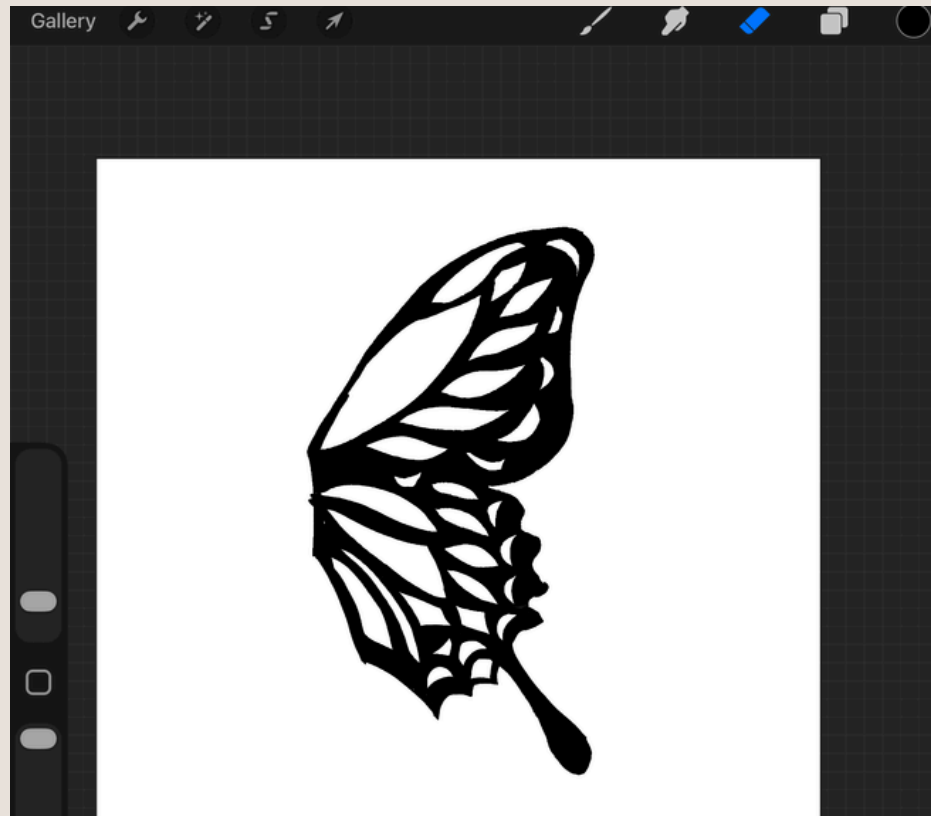


Same base for the other side of the headphones.

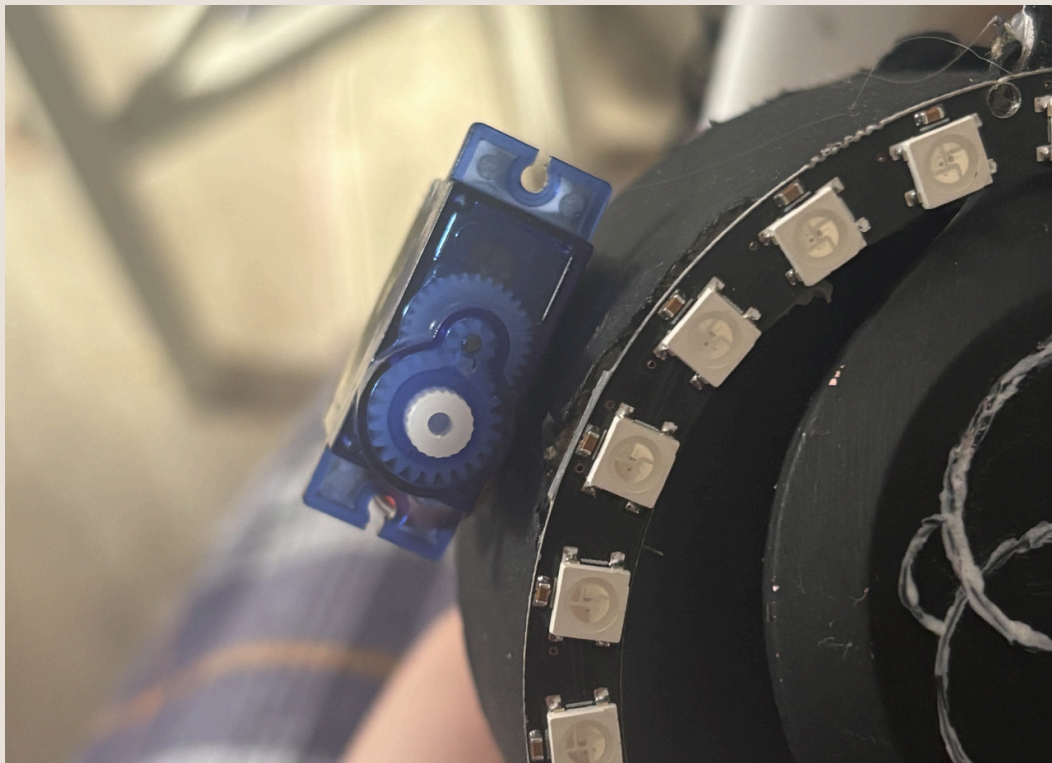


Nano after soldering and hot glue.

PROCESS



I sketched the butterfly on my Ipad then cut it out of cardboard

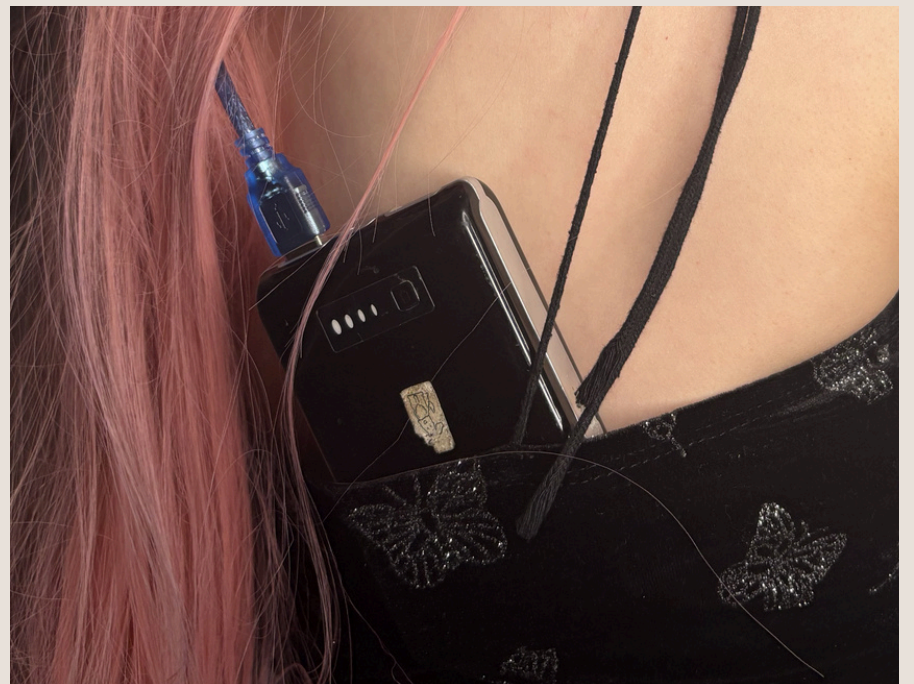
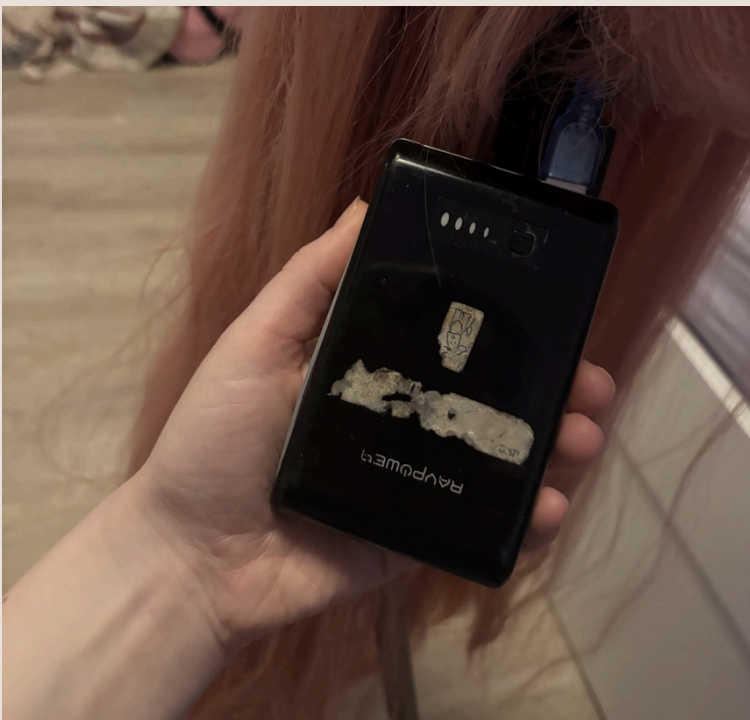


I glued the servo motor onto the side of the headphones.

PROCESS



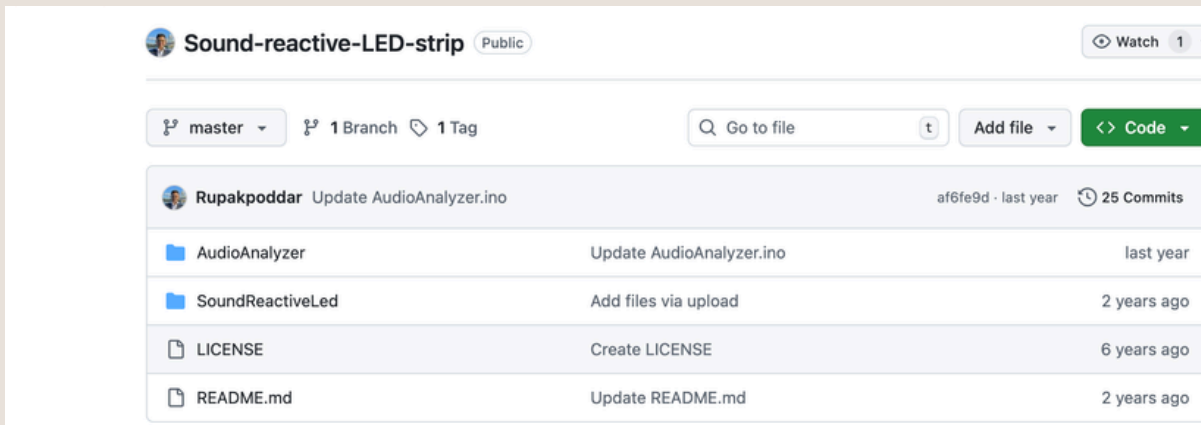
Finished inside (Lots of hot glue)



The headphones are powered by a power bank that can be hidden in the clothes/behind the wig when wearing the headphones.

CODE

<https://github.com/Rupakpoddar/Sound-reactive-LED-strip/tree/master>



As I mentioned earlier, I took most of my code from this GitHub and combined it with my code for the servo. The original code featured a variety of styles for different LED effects, but I decided to only keep “LinearFlowing”, as I think it looks best on the WS2812 ring.

The butterfly servomotor was one of the first things I tested out, so I wrote a code that makes the servo move slightly back and forth. However, when I later tried to combine the servo code with the LED code, I ran into the issue that my servos' delay() was affecting the LEDs. I learned that I had to use millis() instead of delay(). This guide was very enlightening: <https://forum.arduino.cc/t/using-millis-for-timing-a-beginners-guide/483573>, and I had a friend help me with the new servo code!

I ended up with two different Code versions, both of which use the FastLED and Servo library:

Code

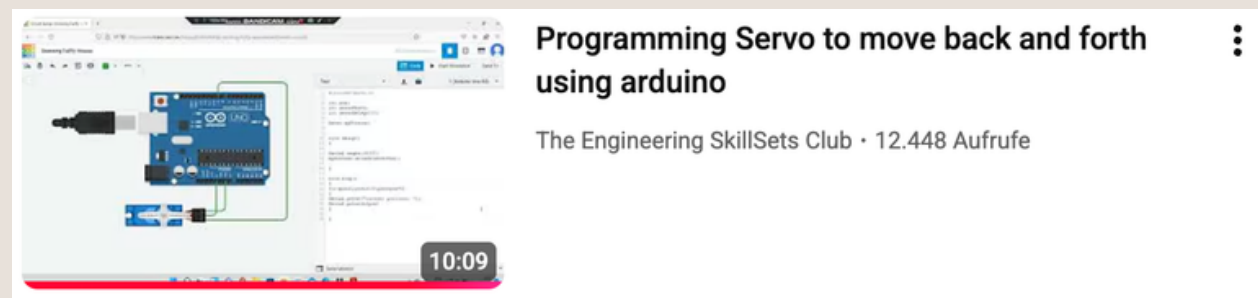
-Colorful LEDs

Code Ver. 2

-Only turquoise

I decided to modify the original code so that the LEDs would only be turquoise (Code Ver. 2), as I think it matches well with the butterfly. Both versions are shown in my video.

Initial servo code:



Very helpful video

```
#include<Servo.h>
```

```
int pos;  
int servoPin=9;  
int servoDelay=30;
```

```
Servo myPointer;
```

```
void setup()  
{  
  Serial.begin(9600);  
  myPointer.attach(servoPin);  
}
```

```
void loop()  
{  
  for(pos=150;pos<=180;pos++)  
  {  
    Serial.print("current position: ");  
    Serial.println(pos);
```

```
    myPointer.write(pos);  
    delay(servoDelay);  
  
  }
```

```
  for(pos=180;pos>=150;pos--)  
  {
```

```
    Serial.print("current position: ");  
    Serial.println(pos);
```

```
    myPointer.write(pos);  
    delay(servoDelay);  
  
  }
```

```
}
```

HARDWARE LIST of the final project

- 1 WS2812 RGB LED Ring
- Arduino Nano
- Microphone Sound Sensor
- Servomotor
- Many wires
- 330 ohm resistor
- Power bank

FINAL REFLECTION (CHALLENGES & POSSIBLE IMPROVEMENTS)

Since I did not follow a tutorial for the hardware, the trickiest part was figuring out what materials to use and how to assemble the headphones. The Arduino Uno was definitely too big, so I'm glad I decided to go for the Nano, as it made hiding it much easier. The soldering was a little tricky as well, but everything worked out fine (except for my first Nano, which I accidentally destroyed by getting solder on a wrong part...).

As long as the music is relatively loud, it can be about half a meter away to still be picked up, which I think is totally sufficient. I wore them for about an hour to take videos and pictures for Social Media, and they worked great.

If I were to make these again, I would like to learn how to 3d model to give the headphones a cleaner look. The microphone slightly picks up the noise from the servo, which I don't mind too much, as playing music overtones the servo. In some videos I took, I could see the blue cable that connects to the power bank. I will probably get a new, longer cable and paint it the color of the wig. Overall, I am really happy with how the headphones came out, and I think they are a great prop for cosplay.

FINISHED HEADPHONES

